

Yejia Liu

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Education	University of California Riverside , Riverside, US <i>Doctor of Philosophy</i> , Computer Science Advisor: Shaolei Ren	2021 - 2025
	Simon Fraser University , Burnaby, Canada <i>Master of Science</i> , Computing Science Advisors: Oliver Schulte, Jiannan Wang	2016 - 2018
	South China University of Technology , Guangzhou, China <i>Bachelor of Engineering</i> , Software Engineering	2012 - 2016
Research Interest	I work on building <i>efficient, equitable</i> and <i>responsible</i> systems across database and machine learning .	
Employment	Data Scientist , Microsoft Redmond, WA, US	Aug 2025 - Present
	Data Scientist Intern , Microsoft Redmond, WA, US	Jun 2023 - Sep 2024
	Software Engineer , Tencent Shenzhen, Guangdong, China	Aug 2019 - Apr 2020
	Software Engineer , Amazon Vancouver, BC, Canada	Sep 2018 - Jun 2019
	Software Engineer Intern , Amazon Vancouver, BC, Canada	May 2017 - Aug 2017
Publications	Wang Bill Zhu*, Miaosen Chai*, Shangshang Wang, Yejia Liu , Song Bian, Honghua Dong, Willie Neiswanger, Robin Jia. Precise Debugging Benchmark: Is Your Model Debugging or Regenerating? <i>Annual Conference of the Association for Computational Linguistics (ACL)</i> , 2026.	
	Shijin Duan, Yejia Liu , Gaowen Liu, Ramana Rao Kompella, Shaolei Ren, Xiaolin Xu. Towards Vector Optimization on Low-Dimensional Vector Symbolic Architecture. <i>Conference on Parsimony and Learning (CPAL)</i> , 2025.	
	Yejia Liu , Pengfei Li, Daniel Wong, Shaolei Ren. Geographical Server Relocation: Opportunities and Challenges. <i>Workshop on Sustainable Computer Systems (HotCarbon)</i> , 2024.	
	Yejia Liu , Jianyi Yang, Pengfei Li, Tongxin Li, Shaolei Ren. Building Socially-Equitable Public Models. <i>International Conference on Machine Learning (ICML)</i> , 2024.	

Pengfei Li*, **Yejia Liu***, Jianyi Yang, Shaolei Ren. **Towards Socially and Environmentally Responsible AI.** *Ethical Systems and Architecture Design Workshop at ASPLOS (HotEthics, ASPLOS)*, 2024.

Yejia Liu, Shijin Duan, Xiaolin Xu, Shaolei Ren. **Scheduled Knowledge Acquisition on Lightweight Vector Symbolic Architectures for Brain-Computer Interfaces.** *EDGE AI Research Symposium (TinyML)*, 2024.

Weiyuan Wu, Pei Wang, Yi Xie, **Yejia Liu**, George Chow, Jiannan Wang. **Web Connector: A Unified API Wrapper to Simplify Web Data Collection.** *Very Large Data Bases (VLDB)*, 2023.

Yejia Liu, Shijin Duan, Xiaolin Xu, Shaolei Ren. **MetaLDC: Meta Learning of Low-Dimensional Computing Classifiers for Fast On-Device Adaption.** *EDGE AI Research Symposium (TinyML)*, 2023.

Yejia Liu*, Wang Zhu*, Shaolei Ren. **Navigating Memory Construction by Global Pseudo-Task Simulation for Continual Learning.** *Conference on Neural Information Processing Systems (NeurIPS)*, 2022.

Shijin Duan, **Yejia Liu**, Shaolei Ren, Xiaolin Xu. **LeHDC: Learning-Based Hyperdimensional Computing Classifier.** *Design Automation Conference (DAC)*, 2022.

Lampros Flokas, Weiyuan Wu, **Yejia Liu**, Jiannan Wang, Nakul Verma, Eugene Wu. **Complaint-Driven Training Data Debugging at Interactive Speeds.** *ACM SIGMOD International Conference on Management of Data (SIGMOD)*, 2022.

Yejia Liu, Weiyuan Wu, Lampros Flokas, Jiannan Wang, Eugene Wu. **Enabling SQL-based Training Data Debugging for Federated Learning.** *Very Large Data Bases (VLDB)*, 2022.

Yejia Liu, Oliver Schulte, Chao Li. **Model Trees for Identifying Exceptional Players in the NHL and NBA Drafts.** *Machine Learning and Data Mining for Sports Analytics Workshop at ECML-PKDD (MLSA, ECML-PKDD)*, 2018.

Teaching

Teaching Assistant, CS 008 (Introduction to Computing) Winter 2025
TA of 250-student class, graded quizzes and assignments. Hold office hours weekly.

Head Teaching Assistant, CS 100 (Software Construction) Fall 2024
Head TA of 130-student class, managed a team of 6 TAs and graders, led weekly discussion sections, proofread all assignments and designed questions for the midterm and final.

Head Teaching Assistant, CS 100 (Software Construction) Spring 2023
Head TA of 130-student class, managed a team of 6 TAs and graders, led weekly discussion sections, proofread all assignments and designed questions for the midterm and final.

Teaching Assistant, CS 008 (Introduction to Computing) Spring 2022
TA of 250-student class, graded quizzes and assignments. Hold office hours weekly.

Teaching Assistant, CS 153 (Design of Operating Systems) Winter 2021

TA of 50-student class, graded quizzes and assignments. Hold office hours weekly.

Teaching Assistant, CS 009P (Introduction to Programming) Fall 2021
TA of 100-student class, led labs, graded quizzes and assignments. Hold office hours weekly.

Mentorship	Shijin Duan (Northeastern University PhD)	Sep 2021 - May 2025
	Zhifeng Wu (UCR PhD)	Sep 2024 - Jun 2026

Professional Services	International Conference on Artificial Intelligence and Statistics (AISTATS 2022–2025)	
	International Conference on Machine Learning (ICML 2022, 2025)	
	Conference on Neural Information Processing Systems (NeurIPS 2025)	

Awards	NeurIPS Scholar Travel Fund	2022
	UCR Graduate School Fellowship	2021
	SFU Graduate School Fellowship	2017
	Mitacs Globalink Research Scholarship	2017
	Women in Computing Science (WiCS) @ SFU	2016 - 2018
	SCUT Outstanding Graduate (Top 1)	2016